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Name: _____

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Department/School: _____

SID: _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu) = \int_0^{+\infty} x^{\nu-1} e^{-x} dx, \quad \Gamma(\nu+1) = \nu\Gamma(\nu), \quad \Gamma(1) = 1, \quad \Gamma(1/2) = \sqrt{\pi}.$$

Select five out of following six problems.

1.(20pts.) In the inertial system S , an event is observed to take place at point A on the x -axis and 10^{-6} s later another event takes place at point B , 900m further down. Find the magnitude and direction of the velocity of S' with respect to S in which these two events appear simultaneous.

Solution: The Lorentz transformation is given by

$$\Delta t' = \gamma \left(\Delta t - \frac{u}{c^2} \Delta x \right).$$

Since as given in the problem, $\Delta t' = 0$, then, we have

$$\Delta t - \frac{u}{c^2} \Delta x = 0.$$

Thus, we obtain

$$u = \frac{\Delta t}{\Delta x} c^2 = \frac{10^{-6} \text{ s}}{900 \text{ m}} \times 3.0 \times 10^8 \text{ m/s} \cdot c = \frac{1}{3} c = 1.0 \times 10^8 \text{ m/s}.$$

2. (20pts.) A pion moving along x -axis with $\beta = 0.8$ in the lab system decays by emitting a muon with $\beta' = 0.268$ along the y' -axis in the rest system of pion. Find the velocity $\mathbf{v}$ of the muon in the lab system.

Solution: The Lorentz inverse velocity transformation is given by

$$\begin{cases} v_x = \frac{v'_x + u}{1 + uv'_x/c^2} \\ v_y = \frac{v'_y}{\gamma(1 + uv'_x/c^2)} \\ v_z = \frac{v'_z}{\gamma(1 + uv'_x/c^2)} \end{cases}.$$

Substituting $u = \beta c = 0.8c$, $v'_x = 0$, $v'_y = \beta' c = 0.268c$, $v'_z = 0$ into above equations, we obtain

$$v_x = \frac{v'_x + u}{1 + uv'_x/c^2} = u = 0.8c,$$

$$v_y = \frac{v'_y}{\gamma(1 + uv'_x/c^2)} = \frac{\beta' c}{\gamma} = \beta' c \sqrt{1 - \beta^2} = 0.161c,$$

$$v_z = \frac{v'_z}{\gamma(1 + uv'_x/c^2)} = 0.$$

3. (20pts.) 3. Consider a monatomic ideal gas of N molecules.

(a) What are the internal energy U and the equation of state of the ideal gas?

(b) Find the entropy S of this gas as the function of T and V , i.e., $S = S(T, V)$, setting (T_0, V_0) as the reference point where the entropy is S_0 .

(c) Find the Helmholtz free energy F of this gas as the function of T and V , i.e., $F = F(T, V)$.

Solution: (a) For a monatomic ideal gas, the isochoric heat capacity is

$$C_V = \frac{3}{2}nR = \frac{3}{2}nN_A k_B = \frac{3}{2}Nk_B$$

Thus,

$$U(T) = C_V T + U_0 = \frac{3}{2}Nk_B T + U_0$$

(The internal energy of an ideal gas is the function of temperature only.)

(Or: By using the equipartition theorem, each molecule of a monatomic ideal gas has 3 degree of freedom. Then, the mean kinetic energy of each molecule is $\frac{3}{2}k_B T$. Thus, the internal energy can be

expressed as $U(T) = \frac{3}{2}Nk_B T + U_0$.)

The equation of state is $pV = nRT = Nk_B T$.

(b) The fundamental equation of thermodynamics is given by

$$dU = TdS - pdV.$$

Then

$$dS = \frac{dU + pdV}{T} = \frac{\frac{3}{2}Nk_B dT + \frac{Nk_B T}{V} dV}{T} = \frac{3Nk_B}{2} \frac{dT}{T} + Nk_B \frac{dV}{V}.$$

Thus, integrating both sides of above equation, we obtain

$$\int_{(T_0, V_0)}^{(T, V)} dS = \int_{(T_0, V_0)}^{(T, V)} \frac{3Nk_B}{2} \frac{dT}{T} + Nk_B \frac{dV}{V},$$

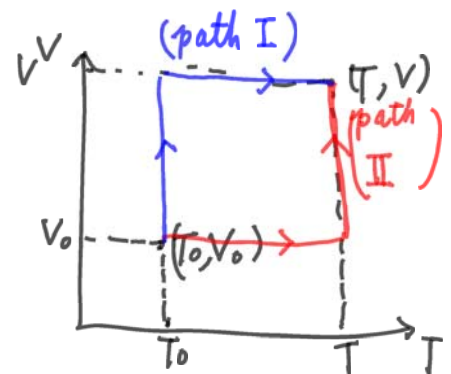
$$S(T, V) - S(T_0, V_0) = \frac{3Nk_B}{2} \ln \frac{T}{T_0} + Nk_B \ln \frac{V}{V_0},$$

$$S(T, V) = \frac{3Nk_B}{2} \ln \frac{T}{T_0} + Nk_B \ln \frac{V}{V_0} + S_0 = Nk_B \ln \left(\frac{T}{T_0} \right)^{\frac{3}{2}} \left(\frac{V}{V_0} \right) + S_0$$

Attention: The above integral is a line integral (curve integral, curvilinear integral). However, since the integrand is an exact differential, the line integral is path independent. This means that you can choose any path that connects the initial state (T_0, V_0) and final state (T, V) . The examples of the paths are path I and path II in the accompanying figure.

(c) The Helmholtz free energy is

$$F = U - TS = Nk_B T \left[1 - \ln \left(\frac{T}{T_0} \right)^{\frac{3}{2}} \left(\frac{V}{V_0} \right) \right] + U_0 - TS_0.$$



4. (20pts.) Consider a gas in an adiabatically enveloped vessel of total volume V_f . The gas is initially at temperature T_i and confined by a divider to a volume V_i . Suppose that a divider is lifted, allowing the gas to abruptly and freely expand into final total volume V_f . At the end of this process, the gas reaches a temperature T_f .

(a) What is the change of the internal energy U in this process?

(b) Show that the temperature change in this process can be expressed as

$$\Delta T = T_f - T_i = - \int_{V_i}^{V_f} \frac{T^2}{C_V} \left(\frac{\partial(p/T)}{\partial T} \right)_V dV.$$

(c) If the gas obeys the Van der Waals equation, calculate the temperature change in this process (the C_V of Van der Waals gas is independent of V). Explain why the temperature change is positive or negative in terms of interaction among the molecules of the gas.

Solution: (a) In the process, no work is done and no heat is transferred. Thus the internal energy is unchanged. $\Delta U = 0$.

(b) To deal with the problem, we consider the quantity $(\partial T / \partial V)_U$. Using the cyclic chain rule, we have

$$\left(\frac{\partial T}{\partial V} \right)_U = - \frac{\left(\frac{\partial U}{\partial V} \right)_T}{\left(\frac{\partial U}{\partial T} \right)_V} = - \frac{\left(\frac{\partial U}{\partial V} \right)_T}{C_V}.$$

From

$$dU = TdS - pdV = T \left(\frac{\partial S}{\partial T} \right)_V + \left[T \left(\frac{\partial S}{\partial V} \right)_T - p \right] dV$$

we have

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial S}{\partial V} \right)_T - p = T \left(\frac{\partial p}{\partial T} \right)_V - p = -T^2 \left(\frac{\partial(p/T)}{\partial T} \right)_V. \quad (dF = -SdT - pdV)$$

(Thus, we have

$$dU = C_V dT - T^2 \left(\frac{\partial(p/T)}{\partial T} \right)_V dV. \quad ***)$$

Thus, writing $T = T(V, U)$ (This can be implied from $U = U(T, V)$) and noting that $dU = 0$ for the process, we have

$$dT = \left(\frac{\partial T}{\partial V} \right)_U dV + \left(\frac{\partial T}{\partial U} \right)_V dU = \left(\frac{\partial T}{\partial V} \right)_U dV = - \frac{T^2}{C_V} \left(\frac{\partial(p/T)}{\partial T} \right)_V dV.$$

(This expression can also be obtained by using *** and letting $dU = 0$.)

Therefore,

$$\Delta T = \int_{T_i}^{T_f} dT = - \int_{V_i}^{V_f} \frac{T^2}{C_V} \left(\frac{\partial(p/T)}{\partial T} \right)_V dV.$$

The Van der Waals equation reads

$$p = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}, \quad \text{or} \quad \frac{p}{T} = \frac{nR}{V - nb} - \frac{n^2 a}{V^2 T}.$$

Then

$$\left(\frac{\partial(p/T)}{\partial T} \right)_V = \frac{n^2 a}{V^2 T^2}.$$

Thus

$$\Delta T = - \int_{V_i}^{V_f} \frac{T^2}{C_V} \left(\frac{\partial(p/T)}{\partial T} \right)_V dV = - \int_{V_i}^{V_f} \frac{n^2 a}{C_V} \frac{1}{V^2} dV = \frac{n^2 a}{C_V} \frac{1}{V} \Big|_{V_i}^{V_f} = \frac{n^2 a}{C_V} \left(\frac{1}{V_f} - \frac{1}{V_i} \right) < 0.$$

The explanation is: As the gas expands, the spacing among molecules increases. Thus the attractive (negative) interaction potential energy increases and the translational kinetic energy of the molecules decreases and therefore the temperature drops.

5. (20 pts.) Consider a rubber band of length l which is being stretched by external force f . Assume that change in volume of the rubber band due to the stretching is negligible.

(a) Write down the fundamental thermodynamic equation for the rubber band, analogous to the fundamental thermodynamic equation $dU = TdS - pdV$ for a system of fixed amount of gas.

(b) Show that $\left(\frac{\partial S}{\partial l}\right)_U = -\frac{f}{T}$.

(c) In an experiment, the length of the band is fixed at $l = 1$ m and the temperature of the rubber band is initially at $T = 300$ K and then is raised by a small increment $\Delta T = 3$ K. This causes the force needed to maintain the length of the band to increase by the amount of $\Delta f = 1.2$ N. In another experiment, the rubber band is stretched from $l = 1$ m by a small amount $\Delta l = 0.02$ m at constant temperature $T = 300$ K. What is the amount of the heat the band exchanges with the environment in the latter experiment? Is the heat released or absorbed by the band?

Solution: (a) The fundamental thermodynamic equation for the rubber band can be written as

$$dU = TdS + fdl.$$

(b) Rewriting the above equation as

$$dS = \frac{1}{T}dU - \frac{f}{T}dl.$$

Since dS is an exact differential, one can see that $\left(\frac{\partial S}{\partial l}\right)_U = -\frac{f}{T}$.

(d) By using the Legendre transformation, the Helmholtz free energy of the rubber band can be written as $F = U - TS$ and its differential is

$$dF = -SdT + fdl.$$

Since it is an exact differential, we have the Maxwell relation $\left(\frac{\partial S}{\partial l}\right)_T = -\left(\frac{\partial f}{\partial T}\right)_l$. Therefore

$$\Delta Q = T\Delta S = T\left(\frac{\partial S}{\partial l}\right)_T \Delta l = -T\left(\frac{\partial f}{\partial T}\right)_l \Delta l = -T \frac{\Delta f}{\Delta T} \Delta l = -300\text{K} \times \frac{1.2\text{N}}{3\text{K}} \times 0.02\text{m} = -2.4\text{J} < 0..$$

The heat is released.

6. (20pts.) 6. The Maxwell velocity distribution describes the probability of a molecule of the ideal gas composed molecules of mass m and in equilibrium of temperature T in the velocity space. It states that the probability of finding a molecule with v_x between v_x and $v_x + dv_x$, v_y between v_y and $v_y + dv_y$, v_z between v_z and $v_z + dv_z$ is given by

$$P(v_x, v_y, v_z)dv_x dv_y dv_z = C \exp\left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right] dv_x dv_y dv_z$$

where C is a normalizing factor.

(a) Determine the normalizing factor C .

(b) Obtain the energy distribution $P(E) dE$, i. e. the probability of finding a molecule of with energy E between E and $E + dE$.

(c) What is the average energy $\langle E \rangle$ of the molecules?

(d) What is the uncertainty of the energy $\Delta E = \sqrt{\langle E^2 \rangle - \langle E \rangle^2}$ of the molecules?

Solution: (a) The probability distribution must be normalized, that is

$$\iiint_{-\infty}^{\infty} P(v_x, v_y, v_z) dv_x dv_y dv_z = 1.$$

Thus

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} C \exp\left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right] dv_x dv_y dv_z = C \int_{-\infty}^{\infty} e^{-\frac{mv_x^2}{2k_B T}} dv_x \cdot \int_{-\infty}^{\infty} e^{-\frac{mv_y^2}{2k_B T}} dv_y \cdot \int_{-\infty}^{\infty} e^{-\frac{mv_z^2}{2k_B T}} dv_z = 1.$$

Since

$$\int_{-\infty}^{\infty} e^{-\frac{mv_x^2}{2k_B T}} dv_x = \sqrt{\frac{2k_B T}{m}} \int_{-\infty}^{\infty} e^{-x^2} dx = 2\sqrt{\frac{2k_B T}{m}} \int_0^{\infty} e^{-x^2} dx = \sqrt{\frac{2k_B T}{m}} \int_0^{\infty} e^{-t} t^{-\frac{1}{2}} dt = \sqrt{\frac{2k_B T}{m}} \Gamma\left(\frac{1}{2}\right) = \sqrt{\frac{2\pi k_B T}{m}}.$$

we obtain

$$C\left(\frac{2\pi k_B T}{m}\right)^{\frac{3}{2}} = 1, \quad C = \left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}}.$$

(b) The Maxwell speed distribution can be written as

$$P(v)dv = \left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} e^{-\frac{mv^2}{2k_B T}} 4\pi v^2 dv.$$

Noting that $E = \frac{1}{2}mv^2$, we have

$$\begin{aligned} P(v)dv &= \left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} e^{-\frac{mv^2}{2k_B T}} 4\pi v \frac{1}{m} \left(d\frac{mv^2}{2}\right) = \left(\frac{1}{2\pi k_B T}\right)^{\frac{3}{2}} e^{-\frac{E}{k_B T}} 4\pi m^{\frac{1}{2}} v dE \\ &= 4\sqrt{2}\pi \left(\frac{1}{2\pi k_B T}\right)^{\frac{3}{2}} e^{-\frac{E}{k_B T}} \sqrt{E} dE = 2\pi \left(\frac{1}{\pi k_B T}\right)^{\frac{3}{2}} e^{-\frac{E}{k_B T}} \sqrt{E} = P(E)dE \end{aligned}$$

(c)

$$\begin{aligned} \langle E \rangle &= \int_0^{\infty} E P(E) dE = \int_0^{\infty} 2\pi \left(\frac{1}{\pi k_B T}\right)^{\frac{3}{2}} e^{-\frac{E}{k_B T}} E^{\frac{3}{2}} dE = \frac{2}{\sqrt{\pi}} k_B T \int_0^{\infty} e^{-t} t^{\frac{3}{2}} dt = \frac{2}{\sqrt{\pi}} k_B T \Gamma\left(\frac{5}{2}\right) \\ &= \frac{2}{\sqrt{\pi}} k_B T \frac{3}{2} \Gamma\left(\frac{3}{2}\right) = \frac{2}{\sqrt{\pi}} k_B T \frac{3}{2} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{3}{2} k_B T. \end{aligned}$$

(d)

$$\begin{aligned} \langle E^2 \rangle &= \int_0^{\infty} E^2 P(E) dE = \int_0^{\infty} 2\pi \left(\frac{1}{\pi k_B T}\right)^{\frac{3}{2}} e^{-\frac{E}{k_B T}} E^{\frac{5}{2}} dE = \frac{2}{\sqrt{\pi}} (k_B T)^2 \int_0^{\infty} e^{-t} t^{\frac{5}{2}} dt = \frac{2}{\sqrt{\pi}} (k_B T)^2 \Gamma\left(\frac{7}{2}\right) \\ &= \frac{2}{\sqrt{\pi}} (k_B T)^2 \frac{5}{2} \Gamma\left(\frac{5}{2}\right) = \frac{2}{\sqrt{\pi}} (k_B T)^2 \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{15}{4} (k_B T)^2. \end{aligned}$$

The uncertainty is

$$\Delta E = \sqrt{\langle E^2 \rangle - \langle E \rangle^2} = \sqrt{\frac{3}{2}} k_B T.$$